



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Conclusion:

The Java program was successfully implemented to demonstrate the use of the super keyword in inheritance. The super keyword was used to access parent class variables, constructors, and methods from the child class, ensuring proper constructor chaining and code reusability. The Employee–Manager and Vehicle Insurance System examples helped in understanding how super simplifies inheritance and supports effective object-oriented programming in Java.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Code:

```
1 class Employee {
2     String name;
3     int id;
4
5     Employee(String name, int id) {
6         this.name = name;
7         this.id = id;
8     }
9 }
10
11 class Manager extends Employee {
12     String department;
13
14     Manager(String name, int id, String department) {
15         super(name, id);
16         this.department = department;
17     }
18
19     void display() {
20         System.out.println("Employee Name: " + super.name);
21         System.out.println("Employee ID: " + super.id);
22         System.out.println("Department: " + department);
23     }
24 }
25
26 class Main {
27     public static void main(String[] args) {
28         Manager m = new Manager("Anurag Thakur", 22198, "Sales");
29
30         m.display();
31     }
32 }
```

Output:

```
Employee Name: Anurag Thakur
Employee ID: 22198
Department: Sales

=== Code Execution Successful ===
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Code:

```

1 class Vehicle {
2     String vehicleNo;
3     String ownerName;
4
5     Vehicle(String vehicleNo, String ownerName) {
6         this.vehicleNo = vehicleNo;
7         this.ownerName = ownerName;
8     }
9 }
10
11 class CarInsurance extends Vehicle {
12     double premium;
13
14     CarInsurance(String vehicleNo, String ownerName, double premium) {
15         super(vehicleNo, ownerName);
16         this.premium = premium;
17     }
18
19     void display() {
20         System.out.println("Vehicle Number: " + super.vehicleNo);
21         System.out.println("Owner Name: " + super.ownerName);
22         System.out.println("Insurance Premium: " + premium);
23     }
24 }
25
26 class Main {
27     public static void main(String[] args) {
28         CarInsurance c = new CarInsurance("MH12AB1234", "Anurag Thakur", 15000);
29
30         c.display();
31     }
32 }

```

Output:

```

Vehicle Number: MH12AB1234
Owner Name: Anurag Thakur
Insurance Premium: 15000.0

=== Code Execution Successful ===

```